Arba Minch University
Institute of Technology
Faculty of Electrical and Computer Engineering
Electronic Communication Engineering Stream

"COMPUTATIONAL METHODS"

(ECEg – 3042)

Compiled by:

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CHAPTER 1

ERROR ANALYSIS AND NUMBER REPRESENTATIONS

MODELING, COMPUTERS and ERROR ANALYSIS

- Numerical methods are techniques by which mathematical problems are formulated so that they can be solved with arithmetic operations.

- Although there are many kinds of numerical methods, they have one common characteristic:

  ✓ They invariably involve large numbers of tedious arithmetic calculations.

- It is little wonder that with the development of fast, efficient digital computers, the role of numerical methods in engineering problem solving has increased dramatically in recent years.

Non – computer Methods

- Beyond providing increased computational firepower, the widespread availability of computers (especially personal computers) and their partnership with numerical methods has had a significant influence on the actual engineering problem – solving process.
- In the pre – computer era there were generally three different ways in which engineers approached problem solving:

1) **Analytical, or Exact, methods.**
2) **Graphical solutions**
3) **Calculators and Slide rules**

Cont …

1. Solutions were derived for some problems using **analytical, or exact, methods**.

   ✓These solutions were often useful and provided excellent insight into the behavior of some systems.

- However, analytical solutions can be derived for only a limited class of problems.
- These include those that can be approximated with linear models and those that have simple geometry and low dimensionality.

Cont …

- Consequently, analytical solutions are of limited practical value because most real problems are nonlinear and involve complex shapes and processes.

2. **Graphical solutions** were used to characterize the behavior of systems.

   ✓These graphical solutions usually took the form of plots.

- Although graphical techniques can often be used to solve complex problems, the results are not very precise/exact.

Cont …

- Furthermore, graphical solutions (without the aid of computers) are extremely tedious and awkward/difficult to implement.
- Finally, graphical techniques are often limited to problems that can be described using three or fewer dimensions.

3. **Calculators and Slide Rules** were used to implement numerical methods manually.

- Although in theory such approaches should be perfectly adequate for solving complex problems, in actuality several difficulties are encountered.

Cont …

- Manual calculations are slow and tedious.
- Furthermore, consistent results are elusive/abstract because of simple blunders/errors that arise when numerous manual tasks are performed.
- During the pre – computer era, significant amounts of energy were expended on the solution technique itself, rather than on problem definition and interpretation
- Today, **computers and numerical methods** provide an alternative for such complicated calculations.

Cont …

- Using computer power to obtain solutions directly, you can approach these calculations without recourse to simplifying assumptions or time-intensive techniques.

Numerical Methods and Engineering Practice

- Since the late 1940s the widespread availability of digital computers has led to a veritable explosion in the use and development of numerical methods.
- There are several reasons why you should study numerical methods:

1) Numerical methods are **extremely powerful problem – solving tools.**

   - They are capable of handling large systems of equations, nonlinearities, and complicated geometries that are not uncommon in engineering practice and that are often impossible to solve analytically.
   - As such, they greatly enhance your problem – solving skills.

Cont …

2) During your careers, you may often have occasion to use commercially available **prepackaged, or "canned," computer programs** that involve numerical methods.

   - The intelligent use of these programs is often based on knowledge of the basic theory underlying the methods.

3) Many problems cannot be approached using canned (pre – recorded) programs. If you are conversant with numerical methods and are adept at computer programming, you **can design your own programs** to solve problems without having to buy or commission expensive software.

Cont …

4) Numerical methods are an **efficient vehicle for learning to use computers.** It is well known that an effective way to learn programming is to actually write computer programs.

5) Numerical methods provide a **vehicle for you to reinforce your understanding of mathematics.** Because one function of numerical methods is to reduce higher mathematics to basic arithmetic operations, they get at the "nuts and bolts" of some otherwise obscure/vague topics.

Approximations and Round – Off Error

- Understanding the concept of error is so important to the effective use of numerical methods.
- Although the numerical technique yielded estimates that were close to the exact analytical solution, there was a discrepancy, or *error,* because the numerical method involved an approximation.
- Actually, we were fortunate in that case because the availability of an analytical solution allowed us to compute the error exactly.
- For many applied engineering problems, we cannot obtain analytical solutions.

Cont …

- Therefore, we cannot compute exactly the errors associated with our numerical methods. In these cases, we must settle for approximations or estimates of the errors.
- In professional practice, errors can be costly and sometimes catastrophic/disastrous.
- Although perfection is a laudable goal, it is rarely, if ever, attained.
- Numerical approximations introduce similar discrepancies into the analysis.
- Again, the question is: How much the next error is present in our calculations and is it tolerable?

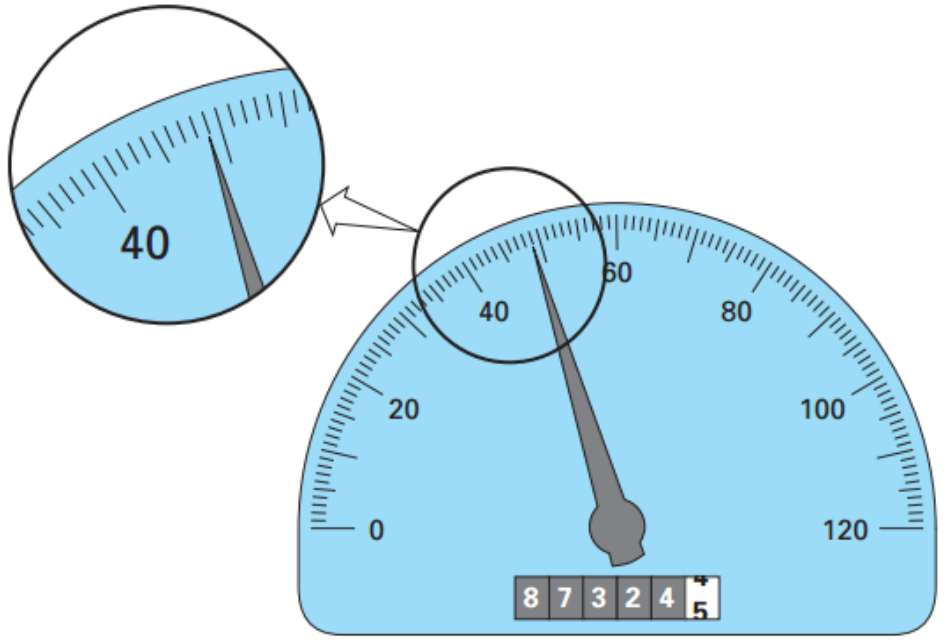
Significant Figures

- Before discussing the errors associated with numerical methods, it is useful to review basic concepts related to approximate representation of the numbers themselves.
- Whenever we employ a number in a computation, we must have assurance that it can be used with confidence.
- For example, Figure below depicts a speedometer and odometer from an automobile.
- Visual inspection of the speedometer indicates that the car is traveling between 48 and 49 km/h.

Cont …

- Because the indicator is higher than the midpoint between the markers on the gauge, we can say with assurance that the car is traveling at approximately 49 km/h.
- We have confidence in this result because two or more reasonable individuals reading this gauge would arrive at the same conclusion.

FIGURE
An automobile speedometer and odometer illustrating the concept of a significant figure.



Cont …

- However, let us say that we insist that the speed be estimated to one decimal place. For this case, one person might say 48.8, whereas another might say 48.9 km/h.
- Therefore, because of the limits of this instrument, only the first two digits can be used with confidence.
- Estimates of the third digit (or higher) must be viewed as approximations.
- It would be ludicrous to claim, on the basis of this speedometer, that the automobile is traveling at 48.8642138 km/h.

Cont …

- In contrast, the odometer provides up to six certain digits.
- From Figure above, we can conclude that the car has traveled slightly less than 87,324.5 km during its lifetime.

- In this case, the seventh digit (and higher) is uncertain.
- The concept of a significant figure, or digit, has been developed to formally designate the reliability of a numerical value.
- The **significant digits** of a number are those that can be used with **confidence**.
- They correspond to the number of **certain digits** plus **one estimated digit**.

Cont …

- For example, the speedometer and the odometer in yield readings of three and seven significant figures, respectively.
- For the speedometer, the two certain digits are 48.
- It is conventional to set the estimated digit at one – half of the smallest scale division on the measurement device.
- Thus the speedometer reading would consist of the three significant figures: 48.5.
- In a similar fashion, the odometer would yield a seven significant – figure reading of 87,324.45.

Cont …

- Although it is usually a straightforward procedure to determine the significant figures of a number, some cases can lead to confusion.

For example:

- Zeros are not always significant figures because they may be necessary just to locate a decimal point.
  - The numbers 0.00001845, 0.0001845, and 0.001845 all have four significant figures.
- Similarly, when trailing/ irregular zeros are used in large numbers, it is not clear how many, if any, of the zeros are significant.

Cont …

- For example, at face value the number 45,300 may have three, four, or five significant digits, depending on whether the zeros are known with confidence.
- Such uncertainty can be resolved by using scientific notation, where $4.53 * 10^4, 4.530 * 10^4, 4.5300 * 10^4$ designate that the number is known to three, four, and five significant figures, respectively.
- The concept of significant figures has two important implications for our study of numerical methods:

1) Numerical methods yield **approximate results**.

Cont …

- We must, therefore, develop criteria to specify how confident we are in our approximate result.
- One way to do this is in terms of significant figures.
  - For example, we might decide that our approximation is acceptable if it is correct to four significant figures.

2) Although quantities such as π, e, or √7 represent specific quantities, they cannot be expressed exactly by a limited number of digits.
   - For example, π = 3.14159265358979323846264 3... ad infinitum.

Cont …

- Because computers retain only a finite number of significant figures, such numbers can never be represented exactly.
- The omission of the remaining significant figures is called round-off error.
- Both round-off error and the use of significant figures to express our confidence in a numerical result will be explored in detail in subsequent sections.
- In addition, the concept of significant figures will have relevance to our definition of accuracy and precision.

Accuracy and Precision

- The errors associated with both calculations and measurements can be characterized with regard to their accuracy and precision.

Accuracy:

- refers to how closely a computed or measured value agrees with the true value.

Precision:

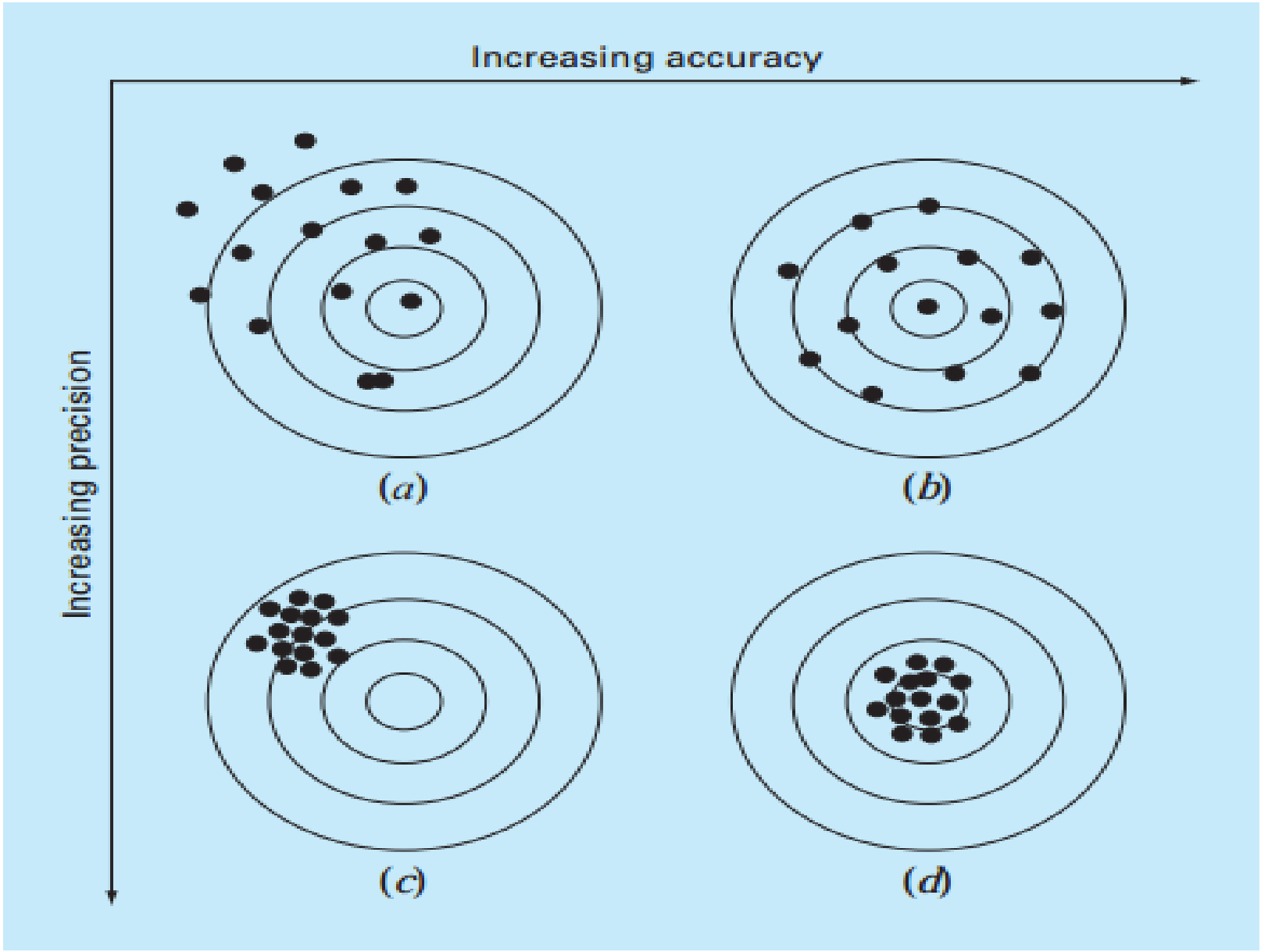
- refers to how closely individual computed or measured values agree with each other.

Cont …

- These concepts can be illustrated graphically using an analogy from target practice.
- The bullet holes on each target in Figure below can be thought of as the predictions of a numerical technique, whereas the bull's – eye represents the truth.
- *Inaccuracy* (also called *bias*) is defined as systematic deviation from the truth.
- Thus, although the shots in Fig. *c* are more tightly grouped than those in Fig. *a*, the two cases are equally biased because they are both centered on the upper left quadrant of the target.

FIGURE

An example from marksmanship illustrating the concepts of accuracy and precision. (*a*) Inaccurate and imprecise; (*b*) accurate and imprecise; (*c*) inaccurate and precise; (*d*) accurate and precise.



Cont …

- *Imprecision* (also called *uncertainty*), on the other hand, refers to the magnitude of the scatter.
- Therefore, although Fig. *b* and *d* are equally accurate (that is, centered on the bull's-eye), the latter is more precise because the shots are tightly grouped.
- Numerical methods should be sufficiently accurate or unbiased to meet the requirements of a particular engineering problem.
- They also should be precise enough for adequate engineering design.

Cont …

- We will use the collective term *error* to represent both the inaccuracy and the imprecision of our predictions.
- With these concepts as background, we can now discuss the factors that contribute to the error of numerical computations.

Error Definitions

- Numerical errors arise from the use of approximations to represent exact mathematical operations and quantities.
- These include **truncation errors**, which result when approximations are used to represent exact mathematical procedures, and **round-off errors**, which result when numbers having limited significant figures are used to represent exact numbers.
- For both types, the relationship between the exact (or true) result and the approximation can be formulated as:

$$\text{True value} = \text{approximation} + \text{error} \qquad (1)$$

Cont …

- By rearranging Eq. (1), we find that the numerical error is equal to the discrepancy between the truth and the approximation, as in:

$$E_t = \text{true value} - \text{approximation} \quad (2)$$

Where: "E_t" is used to designate the exact value of the error.

The subscript 't' is included to designate that this is the "true" error.

- This is in contrast to other cases, as described shortly, where an "approximate" estimate of the error must be employed.
- It takes no account of the order of magnitude of the value under examination..

Cont …

- For example, an error of a centimeter is much more significant if we are measuring a rivet rather than a bridge.

- One way to account for the magnitudes of the quantities being evaluated is to normalize the error to the true value, as in:

$$True\ fractional\ relative\ error = \frac{true\ error}{true\ value}$$

- The relative error can also be multiplied by 100 percent to express it as:

$$\varepsilon_t = \frac{true\ error}{true\ value} * 100\% \qquad (3)$$

Cont …

- Where ε_t designates the true percent relative error.
- Notice that for Eqs. (2) and (3), E and ε are subscripted with a $'t'$ to signify that the error is normalized to the true value.
- For numerical methods, the true value will be known only when we deal with functions that can be solved analytically.
- Such will typically be the case when we investigate the theoretical behavior of a particular technique for simple systems.
- However, in real – world applications, we will obviously not know the true answer a priori.

Cont …

- For these situations, an alternative is to normalize the error using the best available estimate of the true value, that is, to the approximation itself, as in:

$$\varepsilon_a = \frac{approximate\ error}{approximation} * 100\% \qquad (4)$$

- Where the subscript 'a' signifies that the error is normalized to an approximate value.
- Note also that for real – world applications, Eq. (2) cannot be used to calculate the error term for Eq. (4).

Cont …

- One of the challenges of numerical methods is to determine error estimates in the absence of knowledge regarding the true value.
- Certain numerical methods use an iterative approach to compute answers.
- In such an approach, a present approximation is made on the basis of a previous approximation.
- This process is performed repeatedly, or iteratively, to successively compute better and better approximations.

Cont …

- For such cases, the error is often estimated as the difference between previous and current approximations.
- Thus, percent relative error is determined according to:

$$\varepsilon_a = \frac{current\ approximation - previous\ approximation}{current\ approximation} * 100\% \qquad (5)$$

- The signs of Eqs. (2) through (5) may be either positive or negative.
- Often, when performing computations, we may not be concerned with the sign of the error, but we are interested in whether the percent absolute value is lower than a pre-specified percent tolerance, ε_s.

Cont …

- Therefore, it is often useful to employ the absolute value of Eqs. (2) through (5).
- For such cases, the computation is repeated until:

$$|\varepsilon_a| < \varepsilon_s \qquad (6)$$

- If this relationship holds, our result is assumed to be within the pre - specified acceptable level ε_s.
- Note that: we will almost exclusively employ absolute values when we use relative errors.

Round – Off Error

- As mentioned previously, round-off errors originate from the fact that computers retain only a fixed number of significant figures during a calculation.
- Numbers such as π, e, or $\sqrt{7}$ cannot be expressed by a fixed number of significant figures.
- Therefore, they cannot be represented exactly by the computer.
- In addition, because computers use a base – 2 representation, they cannot precisely represent certain exact base – 10 numbers.
- The discrepancy introduced by this omission of significant figures is called round – off error.

END OF CHAPTER ONE

THANK YOU !!!